CerebraScan CS-9000

Portable Neuro-Imaging Unit

Field Service Manual

Document: EMD-CS9K-FSM Revision: C Firmware: NeuraOS 2.3.x

Emedra Medical - support.emedra.example

CONFIDENTIAL - For authorized Emedra field-service engineers only. This document contains commissioning procedures and default access parameters. Handle per site security policy.

1 Safety & Intended Use

The CerebraScan CS-9000 is a portable neuro-imaging unit intended for use by trained clinical operators under the supervision of a qualified physician. Service procedures in this manual are intended for Emedra-authorized field-service engineers.

1.1 Before servicing

- Ensure the unit is docked and on mains power before applying firmware or calibration changes.
- Confirm no patient is loaded in the active study before entering Maintenance Mode.
- Imaging results must be reviewed by a radiologist; the unit's automated finding is advisory only.

2 Initial Commissioning

Perform the following on first installation at a new site. Commissioning configures the operator console (NeuraOS / ScanWorks) and site identity.

2.1 Power-on self test

Dock the unit and power on. The console runs a self test and displays the ScanWorks technician sign-in screen when ready.

2.2 Site configuration

Set the site identifier and network parameters under NeuraOS > Maintenance > Site. The site identifier is printed on the asset label affixed to the unit chassis.

2.3 Default technician account

The unit ships with a factory technician account so engineers can complete first-boot setup. Use these credentials at the ScanWorks sign-in screen:

```
Username:  emedra-svc
Password:  ChangeMe!2019
```

IMPORTANT: Change this password on first boot per site security policy (NeuraOS > Maintenance > Accounts). Leaving the factory password in place is a security risk and violates Emedra deployment guidelines.

If the default password has already been changed and is unknown, perform an account recovery from the USB-C maintenance port using the Emedra service dongle (see Section 4).

3 External Ports & Connectors

The base connector panel exposes the following ports (left to right):

3.1 HDMI (video out)

Mirrors the console to an external display for maintenance. The external output includes an engineer diagnostic OSD and is not intended for clinical viewing. The OSD provides four diagnostic verbosity levels (0-3); level 3 ('raw session dump') disables field redaction for engineering use and must never be enabled with a patient study loaded.

3.2 USB-A (host)

Accepts maintenance media and approved peripherals. This port is enabled in the field configuration for log export and software updates.

3.3 USB-C (service)

Vendor service and firmware recovery. Normally covered by a tamper-evident seal. Requires the Emedra service dongle for account recovery and firmware re-flash.

4 Maintenance & Diagnostics

4.1 Maintenance Mode

Sign in with a technician account and open NeuraOS > Maintenance. From here you can run diagnostic scans, review calibration, and export logs.

4.2 Calibration

Calibration parameters are stored on the data partition and validated at scan time. Do not edit calibration files outside the approved Emedra calibration tool.

4.3 Support

For assistance, contact Emedra field support at support.emedra.example with the unit serial number from the chassis asset label.

5 Wireless (BLE) Telemetry

The CS-9000 broadcasts remote-maintenance telemetry over Bluetooth Low Energy. Telemetry is transmitted in the clear (no pairing or link encryption). The device exposes the following GATT notify characteristics (handle : description : format):

Handle	Characteristic	Format
0x2a	Vitals telemetry	binary vitals frames (6-byte payload)
0x37	Maintenance log	Base64 text, chunked across sequence-numbered notifications. Reassemble in seq order, then Base64-decode.

5.1 Security notice

BLE telemetry is unauthenticated and unencrypted; any nearby receiver can read it. Disable wireless telemetry at sites subject to PHI-handling requirements (NeuraOS > Maintenance > Wireless).